WISP

Wildfire Intelligent Sensing Platform

Full System Architecture Documentation

NASA SBIR Phase I — Nontraditional Aviation Operations for Wildfire Response

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1. Executive Summary

WISP (Wildfire Intelligent Sensing Platform) is an end-to-end drone management and fire prediction platform designed for wildland fire operations. It directly addresses the documented failures of NASA's PAMS TCL 1 field evaluation and extends the state of the art in three areas that PAMS explicitly deferred to future work: mission intelligence, network resilience, and ground crew integration.

PAMS TCL 1 established a baseline: TFR sharing at 100%, OI updates at 87%, and telemetry at 52%. The telemetry failure was caused by a single aerial relay node that did not scale beyond 3 ground nodes. PAMS used a static simulated fire polygon as its fire data. Approval of drone operations was communicated via voice radio. Ground crews were not connected to the system at all.

WISP addresses each of these gaps with a principled architecture:

- Multi-node mesh topology: every drone is a relay node, eliminating the single point of failure that caused 52% telemetry loss.
- ELMFIRE ensemble fire prediction: replaces the static polygon with a probabilistic 60-minute fire spread forecast updated every 20 minutes.
- Adaptive drone targeting: GP-based information field directs drones to locations that maximally reduce fire model uncertainty.
- Digital authorization chain: replaces voice radio approval with one-tap ATGS digital confirmation.
- Ground crew integration: extends the common operating picture to firefighters on the fireline via drone mesh relay.
- Dynamic fleet optimizer: determines optimal drone role allocation based on current fire size, terrain complexity, and available hardware.
- Multi-station federation: multiple ground stations coordinate for large fires, with sector-based domain decomposition and shared boundary observations.

Key Performance Targets vs. PAMS TCL 1 Baseline		
Telemetry delivery:	> 95%	(vs. 52% PAMS TCL 1)
TFR/OI latency:	< 30 sec	(vs. up to 12 min PAMS TCL 1)
Approval workflow:	Digital	(vs. voice radio PAMS TCL 1)
Fire data:	Live ELMFIRE ensemble	(vs. static polygon PAMS TCL 1)
Ground crew coverage:	Full COP	(vs. none PAMS TCL 1)
Prediction cycle:	< 20 min	(new capability, no PAMS baseline)
Offline operation:	Full capability, zero internet dependency	

WISP does not replace PAMS — it builds on it. The WFSS airspace management layer, OI lifecycle, ASTM F3548 compliance, and FREDDIE framework are adopted directly from PAMS.

WISP's contribution is the intelligence, network resilience, and ground crew layers on top of that validated foundation.

2. PAMS TCL 1 Gap Analysis

WISP's design is grounded in the documented failures and explicitly deferred work from NASA's PAMS TCL 1 field evaluation (Wu et al., 2025) and the PAMS GUI paper (Arbab et al., 2025). Each WISP capability traces directly to a documented gap.

PAMS TCL 1 Documented Gap	Source	WISP Response
52% telemetry success rate; single relay node does not scale	Wu et al. Table 2	Multi-node topology: every drone is a relay node; N drones = N-1 redundancy
TFR latency 0–12 minutes; widely distributed	Wu et al. Fig. 9	P0–P4 priority queue with store-and-forward; DTN Bundle Protocol for intermittent links
Approval not implemented digitally; communicated via voice radios	Wu et al. §III.D	Digital authorization chain; ATGS one-tap approval replaces voice coordination
Contingent state removed; lost-link gap acknowledged	Wu et al. §III.D	Pre-declared contingency volume extension activates automatically on nonconformance
Fire line data simulated with a prescribed polygon	Wu et al. §III.C	Live ELMFIRE ensemble; probabilistic 60-min spread forecast updated every 20 min
No mission planning capabilities	Wu et al. §V future work	Full WISP targeting cycle: GP uncertainty, information field, greedy/QUBO selector
No ground crew integration	Neither paper addresses	Ground crew COP: fire perimeter, wind at position, escape routes, safety alerts
Role-based UI modes deferred to future work	Arbab et al. conclusion	IC, ATGS, UAS operator, and ground crew roles with tailored information feeds
Better GCS integration deferred to future work	Wu et al. §V future work	MAVLink-native fleet management; direct OI-to-flight-plan pipeline

3. System Overview

3.1 Architecture

```
┌──────────────────────────────────────────────────────────────────────┐
│                    WISP GROUND STATION (PAMS Case)                   │
│                                                                      │
│  ┌─────────────┐   ┌──────────────┐   ┌────────────────────────────┐ │
│  │  ELMFIRE    │   │  GP / EnKF   │   │   WISP Orchestrator        │ │
│  │ Ensemble    │◄──│ Assimilation │◄──│   20-min cycle             │ │
│  └─────────────┘   └──────────────┘   └────────────────────────────┘ │
│         │ spread ensemble                        │ mission queue     │
│  ┌──────▼────────────────────────────────────────│──────────────────┐ │
│  │      Information Field + Greedy/QUBO Selector │                  │ │
│  └───────────────────────────────────────────────▼──────────────────┘ │
│                              │ OI requests                           │
│  ┌───────────────────────────▼────────────────────────────────────┐  │
│  │              WFSS / UTM Layer  (PAMS-derived)                  │  │
│  │  OI lifecycle | conflict detection | conformance | auth chain │  │
│  └────────────────────────────────────────────────────────────────┘  │
│                              │                                       │
│  ┌───────────────────────────▼────────────────────────────────────┐  │
│  │              Fleet Optimizer                                   │  │
│  │  Role allocation | relay positioning | circumnavigation check │  │
│  └────────────────────────────────────────────────────────────────┘  │
└──────────────────────────────────────────────────────────────────────┘
                               │ cleared flight plans
       ┌───────────────────────▼─────────────────────────────────┐
       │                  DRONE MESH NETWORK                      │
       │  Type 1/2: High-altitude relay + strategic mapping       │
       │  Type 3:   WISP-targeted sensing + local relay           │
       │  [No Type 4: Type 3 handles ground crew comms directly]  │
       └──────────────────────────────────────────────────────────┘
                  │ observations               │ 900 MHz relay
       ┌──────────▼──────────┐   ┌─────────────▼──────────────┐
       │  Back to WISP       │   │  GROUND CREW DEVICES       │
       │  (GP + EnKF update) │   │  Fire perimeter + wind     │
       └─────────────────────┘   │  Escape routes + alerts    │
                                 └────────────────────────────┘
```

3.2 What WISP Does

WISP answers three questions every 20 minutes:

1. Where is the fire model most uncertain about what will happen next?
2. Which of those uncertainties actually affect the predicted fire trajectory?
3. Where should drones fly to resolve the uncertainties that matter?

It then generates drone mission assignments, clears them through the UTM airspace layer, dispatches drones, collects observations, updates the fire model, and repeats. The common operating picture produced — probabilistic fire perimeter, drone positions, ground crew positions, wind field — is shared to all stakeholders including ground crews via the drone mesh.

3.3 What WISP Is Not

WISP does not direct suppression tactics, manage retardant drops, advise on fireline strategy, or move cargo. Every WISP drone function is either sensing/mapping or communications relay. WISP optimizes missions — it does not reduce pilot-to-aircraft ratios or enable single-pilot multi-ship operations. Each drone retains its own PMS 515 compliant authorization chain and licensed operator. This places WISP squarely within the NASA solicitation's accepted scope and outside all four rejection criteria.

4. Fleet Architecture

4.1 Drone Types

WISP eliminates Type 4 drones from the fleet. Type 4 rotorcraft (≤20 min endurance) require continuous swap logistics to maintain persistent coverage — a 12-hour operational period would need 36+ sorties from a single Type 4. Type 3 rotorcraft at ≤2,000 ft AGL provide adequate 900 MHz ground crew link coverage and serve triple duty: WISP sensing target, mesh relay node, and ground crew communications bridge.

Type	Role	Altitude	Endurance	Key Sensors
Type 1 (Fixed-Wing)	Primary relay + strategic perimeter mapping	3,500–8,000 ft AGL	6–14 hrs	EO/IR, Mode C transponder, Silvus radio
Type 2 (Fixed-Wing)	Secondary relay + tactical mapping	3,500–6,000 ft AGL	1–6 hrs	EO/IR, Mode C transponder, Silvus radio
Type 3 (Rotorcraft)	WISP-targeted sensing + local relay + ground crew comms	≤ 2,000 ft AGL	20–60 min	Multispectral (FMC), anemometer, 900 MHz, DoodleLabs

4.2 Fleet Optimizer

The fleet optimizer runs at deployment and on replan triggers. It determines role assignments based on current fire conditions, terrain, available hardware, and marginal information value. It never requires a specific fleet configuration — it works with whatever is available and degrades gracefully.

```
FleetOptimizer inputs:
  available_drones[]          # inventory with type, endurance, sensors, battery
  fire_state                  # ELMFIRE ensemble current output
  terrain                     # DEM for LOS analysis
  ground_crew_positions[]     # nodes requiring connectivity
  atgs_oi_capacity            # max simultaneous OIs the ATGS can manage

Step 1 — Minimum relay count
  Solve LOS facility location over terrain DEM
  Find min R such that all ground nodes have ≤ 2-hop path to ground station
  Assign highest-endurance platforms to relay roles

Step 2 — Circumnavigation check
  if fire_perimeter_length / drone_speed < 0.5 * cycle_time:
      assign 1 Type 3 to circumnavigation pattern
      (covers perimeter continuously; reduces K_sensing needed by 1)
  else:
      all remaining drones follow WISP mission queue

Step 3 — Sensing count from marginal value curve
```

```
  K_sensing = knee of WISP cumulative_gain curve
  (point where adding another drone gives < 5% of first drone's gain)
  K_sensing = min(K_sensing, available_drones - R_min)

Step 4 — Degradation modes
  N=1:  Combined relay + circumnavigation; single-drone WISP mode
  Fixed-wing only:  Adjust observability model D_fmc for altitude
  No GPU:  N=20 ensemble, greedy only, 30-min cycle
  No internet:  Cached HRRR, SA solver replaces D-Wave
```

4.3 Relay-to-Sensing Ratio

Total Drones	Terrain	Relays	Sensing	Notes
3	Flat	1	2	Minimum viable for WISP multi-drone demonstration
4	Flat	1	3	Recommended minimum for Phase I prototype
5	Moderate	2	3	Good for hills / 500-1000m relief
6	Moderate	2	4	Strong sensing coverage
8	Complex	3	5	Mountains / deep canyons
10	Complex	3	7	Large incident, excellent coverage

Relay drones are not passive assets. Type 1/2 at altitude simultaneously provide relay connectivity and run their own sensors (IR perimeter detection, EO observation), contributing observations to the WISP assimilation cycle. The fleet optimizer accounts for these passive observations in the information field.

4.4 Circumnavigation vs. Targeted Sensing

For small fires where a single Type 3 can complete a full orbit within one WISP cycle, circumnavigation is evaluated as a selection strategy alongside greedy and QUBO. The counterfactual evaluator determines which pattern reduces entropy more for the current fire state. Expected behavior: circumnavigation wins on small homogeneous fires (perimeter < 2–3 km), targeted sensing wins on large or terrain-complex fires where uncertainty is spatially concentrated.

5. Multi-Station Federation

For large fires (>10,000 acres) a single ground station cannot meaningfully cover the domain in a 20-minute WISP cycle. Multiple stations deploy with geographic sector assignments. Each runs an independent WISP instance. Stations share boundary observations and fire perimeter data. No single station is authoritative — the architecture is federated, matching the PAMS decentralized design philosophy.

5.1 Sector Coordination

```
StationBroadcast (every cycle, shared across mesh):
  station_id: str
  sector_polygon: GeoJSON         # this station's domain
  fire_perimeter_local: GeoJSON   # current perimeter in sector
  boundary_observations: list     # drone obs near sector edges
  boundary_uncertainty: ndarray   # GP variance at boundary cells
  active_ois: list[OIStatus]      # what drones are doing
  replan_flags: list[str]         # wind shift, blowup potential
  timestamp: datetime

Neighboring stations ingest boundary data into their GP prior.
Observations near sector edges reduce uncertainty on both sides.

GP buffer zone: each station's domain extends 20% into neighbors.
Observations in buffer zone are shared and assimilated by both stations.
```

5.2 IC-Level Common Operating Picture

The Incident Command Post runs an aggregator station (display-only, no drone tasking). All sector stations push their COP data — fire perimeter, drone positions, crew positions, WISP predictions — to the ICP aggregator which renders the full incident picture. Sector boundaries are not fixed — the fleet optimizer recommends rebalancing as the fire evolves asymmetrically.

5.3 Phase Alignment

Single-station WISP is the Phase I deliverable. Multi-station federation is the Phase II extension. The architecture is designed for this from the start — the StationBroadcast interface and GP buffer zone are implemented in Phase I even if only one station is deployed, ensuring Phase II requires integration rather than redesign.

6. WISP Prediction Engine

The prediction engine is the core differentiator from PAMS. Where PAMS used a static prescribed polygon for fire data, WISP runs a continuous cycle of probabilistic fire spread prediction, uncertainty quantification, and adaptive drone targeting.

6.1 ELMFIRE Ensemble

ELMFIRE (Eulerian Level-set Model of FIRE spread) is a GPU-native, physics-based wildfire spread model developed at USFS. It uses Rothermel rate-of-spread physics solved as a level-set PDE, eliminating the grid-direction bias of cellular automata approaches. ELMFIRE has peer-reviewed validation against historical fire events and is used operationally by USFS — this is borrowed evidence that does not require WISP-specific validation for Phase I.

Parameter	Value	Notes
Ensemble size	N = 50 members	Stable variance estimates; increase to 100 for high-complexity terrain
Domain resolution	50 m	Resampled from 30m LANDFIRE
Forecast horizon	60 minutes	Covers 2–3 prediction cycles ahead
GPU	RTX 4090 (24 GB)	Batches N=50 members; ~5 min for full ensemble
Perturbation source	GP posterior variance	Spatially correlated; cells near RAWS get small perturbations
Perturbation method	FFT circulant embedding	O(D log D); avoids $O(D^3)$ full covariance matrix

6.2 Gaussian Process Prior

The GP maintains spatially resolved FMC and wind field estimates with calibrated uncertainty at every grid cell. Wind field treatment uses HRRR forecast as the prior mean with the GP as a correction layer — significantly better than pure GP from sparse RAWS in data-sparse terrain. FMC and wind are both conditioned on RAWS station data, HRRR forecast, and all accumulated drone observations.

The terrain-aware kernel uses elevation difference and aspect difference in addition to geographic distance. Two points on the same aspect and elevation band are more correlated than two points at equal Euclidean distance on opposite sides of a ridge. Hyperparameters: correlation length ~1–2 km for FMC, ~5 km for wind; fitted from RAWS data or physical defaults.

6.3 Information Field

The information field w(x) quantifies the value of a drone observation at each grid cell. It combines GP variance (uncertainty), sensitivity (does this uncertainty affect fire trajectory), and observability (can the drone sensor reliably measure here). The FMC observability model degrades near active fire using a smoke optical depth proxy rather than a simple distance function — more physically grounded than the original IGNIS design.

```
w(x) = GP_var_fmc(x)     * |S_fmc(x)|     * D_fmc(x)
     + GP_var_wind(x)    * |S_wind(x)|    * D_wind(x)
     + GP_var_winddir(x) * |S_winddir(x)| * D_wd(x)

D_fmc(x)  = 0.86 * smoke_clearance(x)   # degrades with smoke optical depth
D_wind(x) = 0.90 * terrain_factor(x)    # degrades in complex terrain

S_v(c) = corr(arrival_times[:, c], perturbation_fields[:, c])
         computed element-wise across N ensemble members
         cost: one matrix operation, milliseconds
```

6.4 Target Selection

Three selection algorithms run in parallel each cycle and are compared on the counterfactual evaluation framework:

- Greedy: iteratively selects highest-value cell, updates GP variance after each selection via closed-form conditional update. Provable (1-1/e) ≈ 63% optimality ratio. Primary operational selector.
- QUBO: encodes the problem as a Quadratic Unconstrained Binary Optimization matrix. Solved via D-Wave QPU (if satellite connectivity available) → simulated annealing → greedy fallback. The comparison between greedy and QUBO is the scientific contribution; quantum hardware is one solver in the chain, not the thesis.
- Circumnavigation: orbit the fire perimeter at constant standoff. Evaluated as baseline for small fires.

The QUBO framing: the claim is not that quantum computers solve this faster — that is not demonstrated at current QPU scale for this problem size. The claim is that the QUBO formulation naturally encodes spatial redundancy in a way greedy cannot globally optimize, and simulated annealing on the ground station GPU provides a strong classical solver for the same formulation.

6.5 Data Assimilation

Two parallel updates after each drone observation batch: GP update (new observations added to conditioning set; posterior variance drops near observed cells) and EnKF update (adjusts each ensemble member's state using Kalman gain with Gaspari-Cohn localization tapering). Replan triggers: >20% variance reduction flags high-information cycle; >30 degree wind deviation triggers immediate partial re-solve.

7. UTM and Airspace Management Layer

The UTM layer is adopted directly from the PAMS/WFSS architecture. WISP's contribution is the WISP-to-WFSS bridge (converting mission requests to OI volumes), the digital authorization chain (replacing voice radio approval), and the contingency volume extension (restoring lost-link handling that PAMS removed).

7.1 OI State Machine

WISP restores the Contingent state that PAMS removed, but redefines it for wildfire operations. Rather than requiring operator initiation, the contingency volume activates automatically when a drone exits its nominal OI volume. The contingency volume is a pre-declared expansion (+200m lateral, +500ft vertical) filed with the OI before activation. This eliminates the lost-link gap acknowledged in the PAMS paper without penalizing routine operational deviations.

```
WFSS OI States (WISP additions in brackets):

  DRAFT → ACCEPTED → ACTIVATED → NONCONFORMING
                                          |
                              [contingency volume activates]
                                          |
                              [CONTINGENT — pre-declared expansion]
                                          |
                              drone returns OR timeout → ENDED

Key changes from PAMS TCL 1:
  1. Accepted→Activated requires digital ATGS approval (not voice radio)
  2. Contingent state restored as auto-activating volume expansion
  3. OI time windows set from WISP expiry_minutes (auto-generated)
  4. Lost-link behavior pre-filed in OI before authorization granted
```

7.2 WISP-to-WFSS Bridge

```
MissionRequest (from WISP):                OI Volume (for WFSS):
  target: (lat, lon)              →          polygon: 200m radius circle
  information_value: float        →          (priority for scheduling)
  dominant_variable: str          →          (logged for post-analysis)
  substitutes: [(lat,lon)]        →          (fallback OI volumes)
  expiry_minutes: float           →          time_window: [now, now+expiry]
  drone_type: UASType             →          altitude_range: per PMS 515
  contingency_volume: Polygon     →          pre-declared expansion volume
  lost_link_behavior: str         →          loiter | return_lrz | descend
```

7.3 PMS 515 Compliance

All PMS 515 requirements are enforced programmatically as hard gates, not checklists. No OI may transition to Activated without the full authorization chain being digitally confirmed: IC approval → airspace authorization type confirmed → NOTAM filed (auto-generated) → TFR deconfliction confirmed → ATGS one-tap digital approval → OI validated by WFSS → drone authorized to launch.

8. Network Architecture

The WISP network directly addresses the root cause of PAMS TCL 1's 52% telemetry failure. That failure was caused by a single aerial relay node that did not scale with ground node count. WISP's every-drone-is-a-relay-node architecture eliminates the single point of failure structurally — adding sensing drones automatically adds relay redundancy.

8.1 Multi-Node Topology

```
PAMS TCL 1 topology (failed at 4 nodes):
  [Ground node] ─┐
  [Ground node] ─┼─── [Single aerial relay] ─── [Ground station]
  [Ground node] ─┘
  Result: 52% telemetry delivery

WISP topology (scales with fleet size):
  [Ground node] ─── [Type 3 drone] ─┐
  [Ground node] ─── [Type 3 drone] ─┼─── [Type 1/2 relay] ─── [Ground station]
  [Ground node] ─── [Type 3 drone] ─┘          │
                                       [Type 1/2 relay 2]
  Every drone = mesh node
  Network failure requires ALL high-altitude drones to fail simultaneously
  Expected delivery: > 95% based on multi-hop mesh literature
```

8.2 Message Priority Queue

PAMS sent telemetry once with no retransmit logic — the primary cause of packet loss. WISP implements a four-tier priority queue with store-and-forward:

Priority	Message Type	Delivery	Max Latency
P0 Emergency	Safety alerts, crew emergency, lost-link	Retransmit every 5s until ACK; all paths	< 500 ms
P1 Safety	Wind shift, blowup potential, ADS-B proximity	Retransmit every 15s until ACK	< 5 sec
P2 Operational	OI transitions, TFR updates, conformance alerts	Periodic rebroadcast at 60s intervals	< 60 sec
P3 Telemetry	Drone position, velocity, sensor readings	Delta-compressed, adaptive rate, store-forward	< 5 min
P4 Background	Fire perimeter updates, WISP cycle reports	Best-effort, dropped if congested	As available

8.3 Telemetry Compression

Three compression strategies eliminate the bandwidth waste that contributed to PAMS's packet loss: trajectory state compression (position + velocity vector; receivers dead-reckon between updates), delta encoding against filed OI (conforming flight requires minimal bandwidth — only deviations transmitted), and adaptive rate control (telemetry rate scales with RSSI/SNR per link pair).

8.4 Delay-Tolerant Networking

Bundle Protocol RFC 9171 provides store-and-forward for intermittent connectivity — designed precisely for environments like wildland fire incidents where a drone flying behind a ridge may lose connectivity for several minutes. OI state transitions and safety alerts are bundled locally and delivered when connectivity resumes. The common operating picture converges regardless of intermittent link outages.

9. Ground Crew Communications

Neither PAMS paper addresses ground crew integration. WISP closes this gap by extending the common operating picture to firefighters on the fireline — the population most at risk and least served by existing digital situational awareness tools. Ground crew devices connect to the nearest Type 3 drone via 900 MHz radio. The Type 3 relays to the ground station via the mesh.

9.1 Information Downlink

Data Element	Source	Frequency
Current fire perimeter	ELMFIRE ensemble mean	Every 20-min cycle
20-min predicted perimeter + uncertainty band	ELMFIRE ensemble spread	Every 20-min cycle
Wind at crew position	GP posterior at crew GPS	Every 20-min cycle
All crew GPS positions	Uplink from all devices	Continuous 30-sec poll
Active aircraft overhead	WFSS ADS-B + OI registry	Continuous
Escape routes and safety zones	Pre-designated + WISP updated	On change
Hazard alerts (wind shift, blowup)	WISP replan trigger	Immediate P0 interrupt
UAS overhead indicator	WFSS conformance monitor	Continuous

9.2 Information Uplink

Ground crew observations feed back into WISP as soft data points — a crew supervisor reporting fire running uphill NE faster than expected triggers a wind direction prior update in that sector and flags a potential WISP replan. Three reporting mechanisms, designed for use with gloves:

- One-tap status: OK / Needs Support / Emergency. Visible to IC and ATGS immediately.
- Fire observation: pre-set options for spotting distance, flame length, rate of spread. Translated to WISP GP soft constraints.
- Free text: logged, not automatically assimilated.

9.3 Safety Alert Protocol

Safety alerts bypass normal message scheduling and transmit simultaneously via all available paths. P0 alerts are retransmitted every 5 seconds until acknowledged. If a crew device goes silent for more than 5 minutes, the ATGS is alerted and the last known position is retained in the COP for 2 hours.

10. Evidence Basis and Validation Plan

WISP currently has architectural arguments grounded in documented PAMS failures and established methods. It does not yet have system-level empirical results. This section distinguishes what is proven, what is borrowed, and what must be demonstrated.

10.1 What Is Already Proven

- The 52% telemetry failure is NASA's own documented result (Wu et al. Table 2). WISP does not need to prove this problem exists.
- ELMFIRE has peer-reviewed validation against historical fire events by USFS. WISP uses it as a physics engine — the fire model credibility is borrowed.
- The greedy selector's (1-1/e) optimality bound for submodular functions is a proven theorem (Krause et al. 2008). No empirical validation needed.
- GP-based adaptive sampling for environmental monitoring has extensive literature support. WISP applies established methods to a new domain.
- Bundle Protocol RFC 9171 is a published IETF standard designed for exactly this connectivity environment.

10.2 What Must Be Demonstrated in Phase I

The minimum viable Phase I evidence package:

4. WISP simulation on historical fire: run ELMFIRE ensemble with GP targeting on 2018 Camp Fire or 2020 Creek Fire data. Show entropy reduction across 5+ cycles. Show that targeted drone observations reduce prediction error more than RAWS-only. This requires no hardware, no flights, no regulatory approval.
5. Network topology simulation: model multi-node vs. single-node mesh under PAMS TCL 1 terrain conditions using ns-3 or EMANE. Quantify expected telemetry delivery improvement. Directly comparable to Wu et al. Table 2.
6. End-to-end cycle time: demonstrate complete WISP cycle (ensemble + targeting + OI generation) completing in under 20 minutes on ground station hardware.

10.3 Phase II Validation

Test	Pass Criterion	Comparison Baseline
Telemetry delivery rate	> 95%	52% PAMS TCL 1 (Wu et al. Table 2)
TFR/OI latency	< 30 seconds median	0–12 min PAMS TCL 1 (Wu et al. Fig. 9-10)
Digital approval chain	100% OIs approved digitally	0% PAMS TCL 1 (voice radio)
Prediction entropy reduction	> 40% after 5 cycles vs. RAWS-only	RAWS-only baseline (no PAMS analog)

Test	Pass Criterion	Comparison Baseline
WISP cycle time	< 20 minutes	N/A (new capability)
Ground crew alert latency	< 60 seconds P0	N/A (new capability)
Fleet optimizer	Valid assignment for N=1 through N=10	N/A (new capability)

11. Software Architecture

11.1 Repository Structure

```
wisp/
├── core/
│   ├── types.py              # TerrainData, GPPrior, EnsembleResult,
│   │                         # InformationField, SelectionResult, MissionRequest
│   ├── config.py             # PMS 515 altitude limits, fuel params, constants
│   └── utils.py              # UTM projection, LOS analysis, distances
├── terrain/
│   └── terrain.py            # LANDFIRE loader, DEM processing
├── prediction/
│   ├── gp.py                 # GP prior, HRRR wind prior, conditional variance
│   ├── elmfire_engine.py     # ELMFIRE ensemble wrapper
│   ├── information.py        # Sensitivity + information field + smoke proxy
│   └── assimilation.py       # EnKF + GP observation update
├── selectors/
│   ├── greedy.py             # Greedy with GP variance update
│   ├── qubo.py               # QUBO construction + D-Wave/SA/greedy fallback
│   ├── circumnavigation.py   # Orbit pattern for small fires
│   └── baselines.py          # Uniform grid baseline
├── utm/
│   ├── wfss.py               # WFSS OI lifecycle (PAMS-derived)
│   ├── bridge.py             # MissionRequest → OI volume conversion
│   ├── airspace.py           # ADS-B, altitude enforcement, conflict detection
│   └── auth_chain.py         # PMS 515 digital authorization gate
├── fleet/
│   ├── optimizer.py          # Role allocation, relay LOS, circumnavigation check
│   ├── allocator.py          # Drone-to-role assignment
│   └── reallocator.py        # Dynamic reallocation on replan triggers
├── network/
│   ├── mesh.py               # Silvus/DoodleLabs interface
│   ├── priority_queue.py     # P0-P4 priority + store-and-forward
│   ├── telemetry.py          # Delta compression, adaptive rate control
│   └── dtn.py                # Bundle Protocol RFC 9171
├── ground_comms/
│   ├── crew_feed.py          # Downlink package assembly
│   ├── crew_uplink.py        # Observation ingestion + GP soft constraint
│   └── alert_manager.py      # P0 safety alert interrupt
├── federation/
│   ├── station_broadcast.py  # Inter-station data sharing
│   ├── boundary_gp.py        # GP buffer zone for sector edges
│   └── aggregator.py         # IC-level COP display (display only)
├── orchestrator.py           # 20-minute cycle coordinator
├── evaluation.py             # Counterfactual comparison framework
└── visualization.py          # Common operating picture display

apps/
├── ground_station/           # ATGS desktop interface
└── crew_app/                 # Ground crew Android application

scripts/
├── run_cycle.py              # Single cycle execution
├── run_comparison.py         # Multi-strategy comparison
└── sim_scenario.py           # Historical fire simulation (Phase I evidence)
```

11.2 Hardware

Component	Specification	Purpose
Compute	Intel i9 / Ryzen 9, 32 GB RAM	WISP orchestrator, GP, EnKF, fleet optimizer
GPU	NVIDIA RTX 4090 (24 GB VRAM)	ELMFIRE ensemble, QUBO SA solver
Storage	2 TB NVMe SSD	LANDFIRE cache, mission archive, telemetry log
Primary radio	Silvus Streamcaster 4200 E+ (S-band)	Primary mesh uplink (same as PAMS)
Backup radio	DoodleLabs (2.4/5.8 GHz)	Backup mesh (same as PAMS)
ADS-B	uAvionix pingStation	Manned aircraft situational awareness
Ground crew link	900 MHz license-free radio	Type 3 to crew device last-mile
Power	AC + 200 Wh LiFePO4 battery	2–4 hrs autonomous; vehicle 12V charging
Case	Pelican 1650 (PAMS-style)	Field portability; IP67

12. Risk Register

Risk	Likelihood	Impact	Mitigation
ELMFIRE ensemble exceeds 20-min cycle on available GPU	Medium	High	Reduce N to 25; extend cycle to 30 min; dual-GPU upgrade path
Phase I lacks empirical results vs. PAMS baseline	High (current state)	High	Run historical fire simulation before submission — this is the top priority action item
D-Wave QPU unavailable (expected)	High	Low	SA on GPU is primary solver; QPU is enhancement only; framed correctly in proposal
FMC accuracy degrades near active fire	High	Medium	Smoke optical depth proxy degrades D_fmc; WISP redirects to non-smoke sectors
Single-pilot multi-ship interpretation risk	Low	High	Maintain explicit one-operator-per-drone architecture; never reduce pilot-to-aircraft ratio
Network improvement claim challenged without field data	Medium	Medium	Network simulation results (ns-3) provide quantitative basis; Phase II field eval confirms
Ground crew 900 MHz LOS to Type 3 at 2000 ft AGL in complex terrain	Medium	Medium	900 MHz range adequate for most terrain; Type 3 can descend to 1500 ft in LOS-limited areas

13. Glossary

Term	Definition
ADS-B	Automatic Dependent Surveillance-Broadcast. Aircraft position broadcast.
ATGS	Air Tactical Group Supervisor. Manages airspace over an incident.
BVLOS	Beyond Visual Line of Sight. UAS operations outside pilot's visual range.
DTN	Delay-Tolerant Networking. Store-and-forward protocol for intermittent links.
ELMFIRE	Eulerian Level-set Model of FIRE spread. GPU-native wildfire spread model (USFS).
EnKF	Ensemble Kalman Filter. Data assimilation method for model state updates.
FMC	Fuel Moisture Content. Key driver of fire spread rate; primary WISP sensing target.
GP	Gaussian Process. Probabilistic model providing spatially continuous uncertainty estimates.
HRRR	High-Resolution Rapid Refresh. NOAA hourly weather forecast model.
IC	Incident Commander. Highest authority at an incident.
LOS	Line of Sight. Unobstructed radio path between two nodes.
LRZ	Launch and Recovery Zone. Designated UAS takeoff and landing area.
NWCG	National Wildfire Coordinating Group. Sets standards for wildland fire operations.
OI	Operational Intent. Declared airspace volume + time window for a drone operation.
PAMS	Portable Airspace Management System. NASA ACERO baseline (TCL 1 validated).
PMS 515	NWCG Standards for Fire Unmanned Aircraft Systems Operations.
QUBO	Quadratic Unconstrained Binary Optimization. Combinatorial optimization formulation.
RAWS	Remote Automated Weather Station. Sparse ground weather network (~50 km spacing).
TFR	Temporary Flight Restriction. FAA-established restricted airspace around an incident.
UTM	UAS Traffic Management. NASA/FAA framework for low-altitude drone operations.
WFSS	Wildland Fire Service Supplier. PAMS adaptation of the UTM USS concept.
WISP	Wildfire Intelligent Sensing Platform. This system.